

SMART CITY · IOT · ARTIFICIAL INTELLIGENCE

IntelliFlow

AI-Powered Cooperative Density-Based Smart Traffic
Management System Using IoT

“Transforming Independent Traffic Signals into an Intelligent Cooperative Network.”

PRESENTED BY

Priyadharshini

Electronics and Communication Engineering (ECE)



THE PROBLEM

Independent Signals, A Networked Problem

Urban traffic congestion is rising fast as vehicle numbers grow — and today's signal systems aren't built to keep up.

CURRENT CHALLENGES

- 1 Operate independently, without communicating with nearby intersections
- 2 Allocate green time using fixed timers or local vehicle density only
- 3 React only after congestion has already formed
- 4 Shift traffic from one intersection to another instead of eliminating it
- 5 Lack centralized real-time monitoring and predictive decision-making

IMPACT

What Congestion Actually Costs

-  Increased travel time
-  Fuel wastage
-  Driver frustration
-  Traffic bottlenecks
-  Delayed emergency response

THE SOLUTION

IntelliFlow: One Cooperative Traffic Network

An IoT-enabled, density-based traffic management system where multiple intersections work together — instead of operating in isolation.



Real-Time Density Monitoring

Continuous sensing of vehicle count and queue length



Multi-Intersection Communication

Neighboring junctions exchange live traffic data



AI-Assisted Prediction

Forecasts congestion before it builds up



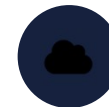
Adaptive Green Allocation

Signal timing tuned to real traffic, not fixed timers



Emergency Priority Override

Temporary green corridor for emergency vehicles



Cloud Monitoring Dashboard

Live analytics and control from anywhere

Unlike conventional systems, IntelliFlow optimizes the entire traffic corridor — not just a single junction.

OBJECTIVES

What IntelliFlow Is Built to Achieve



Reduce traffic congestion across connected junctions



Improve traffic flow across the entire corridor



Optimize green signal duration using real vehicle density



Support fast, prioritized emergency vehicle movement



Enable communication between neighboring intersections



Build a scalable IoT-based Smart City solution



Predict congestion before it becomes severe

SYSTEM ARCHITECTURE

From Road Sensors to Signal Decisions



IOT ARCHITECTURE

Five Layers, One Coordinated System



Sensing Layer

Ultrasonic & IR sensors

Vehicle count · Queue length · Waiting time



Edge Layer

ESP32 on-node processing

Density calculation · Queue analysis · Local decisions



Communication Layer

ESP-NOW + MQTT

Junction-to-junction link · Cloud connectivity



Cloud Layer

Firebase

Live & historical data · AI predictions · Signal status



Application Layer

Dashboard

Monitoring · Analytics · Reports · Recommendations

WORKING METHODOLOGY

How IntelliFlow Responds, Step by Step



 Emergency Priority Mode steps

BUILD STACK

Hardware & Software



Hardware

- ESP32 microcontroller
- Ultrasonic sensors
- IR sensors
- Traffic signal LEDs
- Breadboard
- Jumper wires
- Power supply



Software

- Arduino IDE
- Python
- Firebase
- MQTT broker
- ESP-NOW protocol
- VS Code
- GitHub

Two Systems, One Decision Loop



Role of IoT

- Real-time sensing
- Data collection
- Communication
- Cloud connectivity



Role of AI

- Congestion prediction
- Signal timing optimization
- Corridor-level coordination

AI does not directly control traffic signals — it generates optimized recommendations, while the ESP32 controller safely executes them.

INNOVATION

What Sets IntelliFlow Apart



01

Cooperative Multi-Intersection Communication

Signals exchange traffic information instead of working independently.



02

Predictive Traffic Management

AI predicts congestion before it develops, not after.



03

Network Recovery Mode

Coordinates upstream and downstream intersections to stop congestion spreading.



04

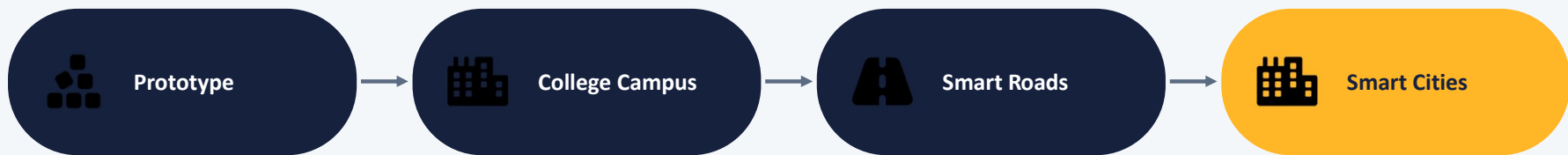
Emergency Priority Override

Temporary priority for ambulances, fire trucks, and police — then reverts automatically.

FEASIBILITY & OUTCOMES

Practical Today, Scalable Tomorrow

SCALABILITY PATH



TECHNICAL FEASIBILITY

Built on affordable, widely available components — no expensive infrastructure required for the prototype.

ESP32

Ultrasonic Sensors

Wi-Fi

MQTT

Firebase

EXPECTED OUTCOMES

- ✓ Reduced waiting time
- ✓ Improved traffic flow
- ✓ Faster emergency response
- ✓ Better inter-junction coordination
- ✓ Centralized traffic monitoring
- ✓ Low-cost deployment

FUTURE SCOPE

Where IntelliFlow Goes Next



Camera-based vehicle classification



Integration with navigation systems



Accident detection



Edge AI deployment



Smart City integration



Digital Twin simulation

CONCLUSION

IntelliFlow combines IoT, AI, cloud computing, and cooperative communication to turn independent traffic signals into an intelligent, connected network — improving traffic efficiency while staying scalable, affordable, and smart-city ready.